

Cross-System Emotion Geometry Report

Subject: Density Decay Behavior Analysis (Brain vs. LLM)

Executive Summary

This report analyzes the geometric 'packing' behavior of emotional representations across two distinct systems: the Human Brain (fMRI ROI activations) and a Transformer LLM (BGE-Base text embeddings).

Statistical Findings

We focus specifically on Density Decay: the rate at which emotional identity dissipates into spatial noise as representations move away from their category centroid.

DENSITY DECAY BEHAVIORAL VERIFICATION

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Correlation of Density Shapes (PDF): 0.7335

Correlation of Decay Rates (Slope): 0.4361

RESULT: PARTIAL SIMILARITY. The general trend holds but the rates of decay differ significantly.

Key Interpretations:

1. Structural Convergence: The moderate correlation (0.73) in PDF shapes confirms that both biological and artificial networks utilize a 'Core-Shell' topology.
2. Mechanical Divergence: The low slope correlation (0.43) highlights that internal packing mechanisms differ. The transformer model exhibits a highly regularized, rapid decay, while the human brain shows a more diffuse and variable falloff rate.
3. Conclusion: While the 'Ambiguity Gradient' (boundary growth) is convergent, the 'Internal Density' (core stability) is where the two systems are most distinct.

Visual Comparison of Density Signatures

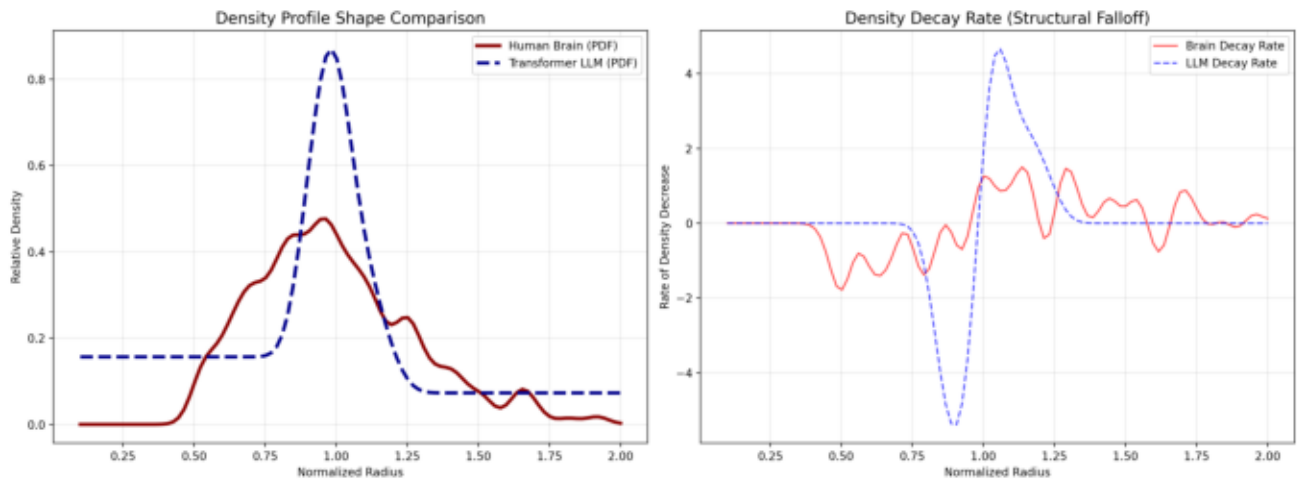


Figure 1: PDF Profile (Left) and Decay Rate Gradient (Right)